

Practice Quiz: Cognition in Robotic Systems

Part A: Multiple Choice

1. In a robotic system, cognition is best understood as: A) The physical hardware of the processor B) The computational processes that maintain, update, and integrate the generative model over time C) The robot's ability to communicate with humans D) The electrical current flowing through circuits

Answer: B -- Cognition in Active Inference encompasses the internal computational processes that maintain beliefs about the world state, integrate information across time and sensory modalities, and update the generative model. It bridges perception (incoming data) and action (outgoing commands).

1. A mobile robot maintains a costmap that integrates lidar scans over time to build an obstacle representation. This costmap is best described as: A) A sensory state of the Markov blanket B) An active state of the Markov blanket C) An internal state -- a cognitive representation maintained by the robot's generative model D) An external state belonging to the environment

Answer: C -- The costmap is an internal belief state that the robot maintains and updates through cognitive processes. It integrates multiple perceptual inputs over time into a coherent spatial representation used for planning and decision-making.

1. Working memory in a robotic system corresponds to: A) The robot's hard drive storage capacity B) Actively maintained belief states that are updated in real time and influence ongoing action selection C) Data backed up to cloud storage D) The total number of sensors on the robot

Answer: B -- Working memory in Active Inference terms consists of actively maintained posterior beliefs about current and recent states that the robot uses for ongoing decision-making -- such as tracking the positions of multiple objects during a manipulation task.

1. What role does precision weighting play in robotic cognition? A) It determines the physical weight of the robot B) It modulates which sensory channels and internal predictions have the most influence on belief updates, functioning as a form of attention C) It sets the speed of the robot's motors D) It controls the robot's power consumption

Answer: B -- Precision weighting in Active Inference acts as an attention mechanism. A robot handling a delicate object increases the precision (weighting) of its force sensor predictions relative to visual predictions, focusing cognitive resources on the most task-relevant information channel.

1. A drone performing visual SLAM simultaneously estimates its own pose and builds a 3D map. The cognitive challenge here is: A) Choosing the right color for the drone B) Jointly estimating interdependent hidden states (pose and map) from ambiguous observations, requiring sophisticated approximate inference C) Flying as fast as possible D) Minimizing the weight of the drone

Answer: B -- SLAM is a canonical example of robotic cognition: the robot must jointly infer its own position (which depends on the map) and the map structure (which depends on the position), resolving circular dependencies through iterative approximate inference methods.

1. Hierarchical generative models in robotic cognition allow: A) Only single-level reasoning about immediate sensor readings B) Reasoning at multiple abstraction levels -- from low-level sensor processing to mid-level object recognition to high-level task understanding C) Eliminating the need for any sensors D) Reducing the robot to a single computational process

Answer: B -- Hierarchical models enable a manipulation robot to simultaneously reason about joint torques (low level), object poses (mid level), and task goals like "assemble the widget" (high level). Each level predicts and constrains the level below.

1. When a robot encounters a situation it has never seen before, high prediction error across all levels of the generative model indicates: A) The robot is functioning perfectly

B) Cognitive surprise -- requiring either significant model updating, exploratory action, or a request for human assistance C) The robot should ignore the novel situation D) All sensors have failed simultaneously

Answer: B -- Pervasive high prediction error signals genuine novelty. The robot must decide whether to update its model (accommodate the new situation), act to reduce uncertainty (explore), or escalate to a human operator if the novelty exceeds its adaptive capacity.

1. A robot performing a multi-step assembly task must maintain a belief about which steps have been completed. This is an example of: A) Passive perception B) Motor control C) Cognitive state tracking -- maintaining a belief state over a task-level generative model with discrete state transitions D) Sensor calibration

Answer: C -- Task state tracking requires the robot to maintain and update beliefs about a hidden variable (current task state) that evolves through discrete transitions (step completed, step failed). This is a core cognitive function that operates at a higher abstraction level than sensorimotor control.

1. The concept of "counterfactual reasoning" in robotic cognition refers to: A) Counting the number of facts in the robot's database B) Mentally simulating the consequences of actions not yet taken, evaluating alternative futures through the generative model C) Reasoning only about past events D) Rejecting all factual information

Answer: B -- An Active Inference robot uses its generative model to simulate "what would happen if I did X versus Y" -- evaluating expected free energy under different action policies before committing to one. This counterfactual evaluation is central to deliberative planning.

1. Belief propagation in a factor graph used for robot state estimation is, from a cognitive perspective: A) Unrelated to cognition B) A distributed inference algorithm where local computations at each node integrate information from neighboring nodes, implementing cognitive integration across the generative model C) Only useful for social media robots D) A method for physical object transport

Answer: B -- Factor graph belief propagation implements cognitive integration: each variable node (pose, landmark, calibration parameter) updates its belief by combining messages from

connected factor nodes (sensor measurements, dynamics constraints), achieving global coherence through local computation.

Part B: Short Answer and Design Prompts

1. A robot chef must track the state of multiple dishes being cooked simultaneously, each at different stages of preparation. Design a cognitive architecture using Active Inference that maintains beliefs about each dish's state, detects when a dish needs attention (prediction error), and allocates the robot's actions appropriately.
2. Explain how a quadruped robot like Boston Dynamics' Spot uses hierarchical cognition: low-level leg coordination, mid-level terrain assessment, and high-level mission planning. How do prediction errors propagate between levels?
3. Compare the cognitive demands of a robot performing repetitive pick-and-place in a structured factory versus a robot navigating a disaster site for search and rescue. How do the generative model requirements differ?
4. How would you implement an "attention mechanism" for a mobile manipulator that must handle both navigation hazards and manipulation targets? Describe how precision weighting would shift between sensory channels as the robot transitions from traveling to grasping.
5. A collaborative robot must infer the intention of a human coworker from observed movements. Design a cognitive model that maintains beliefs about the human's goal and adapts the robot's actions to complement the human's plan. What are the key hidden states and observation models?